Assignment 2
Black-box testing

Assignment Objectives:

- Generating test cases based on black box testing.
- Use JUnit for implementing the test cases.
- Use Jenkins for executing the test cases

Prerequisite:

- Create a team –5 or 6 members.

Assignment = In-Class assignment + Take-Home assignment

Before Lab2: Docker-Jenkins-Gitea–AllureReport - install on your computer

See ***Info_CI_CD.pdf* tutorial.**

1. **In-Class assignment**
   - First hour of the laboratory.
   - Maxim 50 XP
   - TO DO:
     1. **Design test cases (BBT)** (20 XP)
        - Design test cases based on the specification for the given problem.
        - Execute the test case and document the actual results versus expected results.
        - Use the template document: Lab02_BBT_Form.xls.
     2. **BBT Tests** (20 XP)
        - Create a Maven project. Use the Tutorial from Lecture 2.
        - Implement your solution to the given problem.
        - Implement the test cases that you have designed and test your implementation.
     3. **[CI_CD]** (10 XP)
        - Add the project to git using gitea. Use the Tutorial from Lecture 2.
        - Creating a pipeline job in Jenkins for executing the test cases.
   - In-class Problem: Print the number of even numbers in an array of n natural elements.
     1. Data: noElementsArray, valuesArray
     2. Result: noOfEvenElements
     3. Read from file IN.TXT. Print on console.
     4. Run: …>java -jar EvenNumbers.jar

2. **Take-Home assignment**
   - At home.
   - Maxim 100 XP
   - TO DO:
     1. **Design test cases (BBT)** (30 XP)
        - Design test cases based on the specification for the given problem.
     2. **BBT Tests** (30 XP)
        - Create a Maven project. Use the Tutorial from Lecture 2.
        - Implement your solution to the given problem.
        - Implement the test cases that you have designed and test your implementation.
     3. **[CI_CD]** (40 XP)
        - Add the project to git using gitea. Use the Tutorial from Lecture 2.
        - Creating a pipeline job in Jenkins for executing the test cases.
   - Take-Home Problem:The method *GiveBonus* increases the salary of some employees with 1000 lei if the department of the employee had the biggest sale. If the salary of an eligible employee is greater than 5000 lei or if he/she is a manager then the salary will be increased only by 500 lei.
     1. Data: numberEmployees, listEmployees (each employee is described by *name, department, function, salary*)
        numberSales, listSales (each sale is described by *department, sumSale*)

2. Result:
   - If the method computes correctly the requirement, the return code is 0.
   - If either the numberEmployees of the numberSales is 0, the return code is 1.
   - If there are no employees in the department with the biggest sale, the return code is 2.
3. Read from file IN.TXT. Print in file OUT.TXT
4. Run: …>java -jar GiveBonus.jar

Assignment and Delivery date:

1. Assignment date: laboratory 2
2. Delivery date for **In-Class assignment:** laboratory 2 (max 50 XP)
3. Delivery date (first) **for Take-Home assignment**: laboratory 3 (max 100 XP)
4. Delivery date (last) **for Take-Home assignment**: laboratory 4 (max 50 XP)

Turn in:

Delivered and presented in class AND upload on Teams (only one student from the team) after Delivery in class the following archive:

a. Lab2_IC_TH_93X_Name01Name02Name03Name04Name05.zip
   i. In-class problem
      1. Lab02_BBT_Form.xls
      2. Source code: Implementation of the test cases.
      3. Screen capture with the Gitea and Jenkins and Allure report of your tests.
   ii. Take-Home problem
      1. Lab02_BBT_Form.xls
      2. Source code: Implementation of the test cases.
      3. Screen capture with the Gitea and Jenkins and Allure report of your tests.
   iii. document with the list of members. No "guest name"! Every person mentioned as a collaborator must contribute.

References

- See Lecture 2, Seminar 2.
- See documents for Lab02Files.zip